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AGORA: from public data to grounded answers

A standalone research prototype designed and built by Salman Youssef.

Plan
Decompose the question

→

Retrieve
Hybrid catalogue search

→

Route
Select datasets and tools

→

Execute
Retrieve or analyse data

→

Synthesise
Answer with sources

Two levels of evidence

Search metadata across 73,000+ French open-data catalogue records, then reason over the selected datasets.

Engineering contribution

FastAPI services, DSPy orchestration, Weaviate retrieval, streamed progress, provenance, and per-stage token/cost tracking.

Why this design matters

Catalogue relevance does not guarantee a usable resource. The workflow exposes download failures and preserves source links so the user can inspect the evidence.

AGORA research manuscript and local-PC demonstration | Prototype implementation and preliminary evaluation.

What the evidence supports

Implementation evidence, preliminary research findings, and proposed improvements.

Preliminary evaluation

Eight discovery and analytical questions explored planning, retrieval/routing behaviour, answers, and resource failures.

Model comparison limits

GPT-5-mini and LLaMA-3.3-70B-Instruct were explored with different retrieval settings. This is qualitative evidence, not a controlled ranking.

Local-PC demo timing

The featured query took approximately 2 min 30 sec. The published 82-second video contains labelled acceleration. Selected resources are downloaded on demand.

Next research and engineering steps

Ground-truth evaluation, controlled ablations, parallel independent analyses, caching, selective API access, and measured latency/quality trade-offs.

Code: github.com/salmanyoussef/agora

AGORA research manuscript | Production optimisations are proposed; their speedup remains to be measured.